CREATE SCHEMA AND MODELS

Introduction

In the Online Complaint Registration System, MongoDB is used as the NoSQL database to store application data. Mongoose acts as an Object Data Modeling (ODM) library, allowing the application to define schemas and models for interacting with the database.

A **Schema** defines the structure of a document, while a **Model** is created from the schema and is used to perform Create, Read, Update, and Delete (CRUD) operations.

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1. User Schema (User.js)

The User model stores information about registered users and administrators.

Field	Data Type	Description
name	String	User's full name
email	String	User email (unique)
phone	String	Mobile number
password	String	Encrypted password
role	String	User/Admin
createdAt	Date	Registration date

Sample code

```
const mongoose = require("mongoose");

const userSchema = new mongoose.Schema({
   name:{
      type:String,
```

```
        required:true
    },
    email:{
        type:String,
        required:true,
        unique:true
    },
    phone:{
        type:String,
        required:true
    },
    password:{
        type:String,
        required:true
    },
    role:{
        type:String,
        default:"User"
    }
},{
    timestamps:true
});

module.exports = mongoose.model("User", userSchema);
```

Complaint Schema (Complaint.js)

The Complaint model stores all complaints submitted by users.

Field	Data Type	Description
userId	ObjectId	Reference to User
department	String	Department name
category	String	Complaint category
title	String	Complaint title
description	String	Complaint details

location	String	Complaint location
image	String	Uploaded image
status	String	Pending/In Progress/Resolved
createdAt	Date	Complaint date

Sample code

```
const mongoose = require("mongoose");

const complaintSchema = new mongoose.Schema({

    userId:{
        type:mongoose.Schema.Types.ObjectId,
        ref:"User",
        required:true
    },

    department:{
        type:String,
        required:true
    },

    category:{
        type:String,
        required:true
    },

    title:{
        type:String,
        required:true
    },

    description:{
        type:String,
        required:true
    },

    location:{
        type:String,
        required:true
```

```
    },

    image:{
        type:String
    },

    status:{
        type:String,
        default:"Pending"
    }

},{
    timestamps:true
});

module.exports = mongoose.model("Complaint", complaintSchema);
```

Department Schema (Department.js)

Stores available government departments.

Sample code

```
const mongoose = require("mongoose");

const departmentSchema = new mongoose.Schema({

    departmentName:{
        type:String,
        required:true
    },

    description:{
        type:String
    }

},{
    timestamps:true
});
```

```
module.exports = mongoose.model("Department", departmentSchema);
```

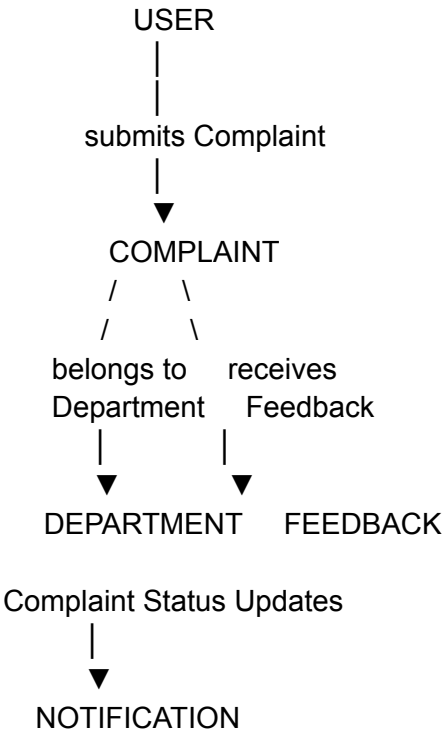
Notification Schema (Notification.js)

Stores notifications sent to users regarding complaint updates.

Sample code

```
const mongoose = require("mongoose");

const notificationSchema = new mongoose.Schema({

    userId:{
        type:mongoose.Schema.Types.ObjectId,
        ref:"User"
    },

    message:{
        type:String,
        required:true
    },

    isRead:{
        type:Boolean,
        default:false
    }

},{
    timestamps:true
});

module.exports = mongoose.model("Notification", notificationSchema);
```

Schema Relationship



Summary

The Online Complaint Registration System uses **Mongoose schemas** to organize and validate application data. Each schema represents a specific entity such as **User**, **Complaint**, **Department**, **Feedback**, and **Notification**. These models ensure data consistency, simplify CRUD operations, and establish relationships between collections using MongoDB references. This modular structure makes the application scalable, maintainable, and easy to extend with additional features in the future.